

Yulia Kumar, Ph.D.

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SUMMARY

A Professional Educator with 10+ years of expertise in teaching a broad range of Computer Science (CS) courses. Pursuing their second PhD in Electrical and Computer Engineering – Machine Learning from Rutgers University. They deeply instruct and mentor students in CS and advanced AI topics, specifically in Large Language Models, Transformers, and Multimodal systems.

RESEARCH INTERESTS: Security, privacy, and trustworthiness of AI-enabled software; LLM and agentic-system security; prompt injection; agent jacking; data and retrieval poisoning; secure retrieval-augmented generation (RAG); LLM firewalls; least-privilege tool authorization; runtime monitoring; human-in-the-loop safeguards; multimodal AI; and distributed AI systems.

EXPERIENCE

- Sep 2019 - Present
- KEAN UNIVERSITY | UNION, NJ**
Senior Lecturer (Department of Computer Science and Technology)
- Expert in Java, Python, JavaScript, C#, HTML/CSS/JS/Node.JS/JS Frameworks, Agentic AI, AI apps.
 - Teaches undergraduate courses Transition to Kean, Fundamentals of Computer Science, AI Exploration, Computer Programming, Computer Organization and Architecture, Data Structures, Software Development with Frameworks, Python Programming, Programming World Wide Web Server, Web Client-Side Programming, .NET Framework and C#, Independent Studies / Senior Research in CS/IT.
 - Teaches graduate courses Web. Design and Development and Data Visualization, mentors graduate independent studies in CS/CIS; Mentors students in AI research, including NLP, Recommender Systems, Computer Vision, Transformer NNs, LLMs, Multimodal and Distributed ML.
 - Develops and teaches secure agentic systems across undergraduate and graduate computing courses, including context engineering, RAG, tool use, agent orchestration, multi-agent workflows, loop and harness engineering, LLM firewalls, observability, prompt-injection defenses, agentjacking mitigation, and human oversight.
- Sep 2014 - Dec 2015
- Academic Specialist (Office of Institutional Research)**
- Specialist in Visual Basic, .NET, SQL server, HTML/CSS/JS, Google Analytics
 - Supported the .NET website and participated in its upgrade to a more relevant .NET platform.
 - Maintained SQL server databases, developed reports using Visual Basic, and performed data mining.
- Sep 2013 - Aug 2014
- Math Tutor (Department of Mathematics)**
- Taught Pre-calculus, Calculus I, II, III, Matrix algebra, Applied Statistics I, II to undergraduates.
- Jun 2014 - Aug 2014
- Math Tutor/Mentor (Upward Bound Project)**
- Taught Pre-calculus, Calculus I, and Algebra II to high school students.
- May 2019 - Aug 2019
- NELSON LABS | FAIRFIELD, NJ**
Software Developer (Gibraltar Laboratories, IT Department)
- Expert in C#, Java, SQL server, PowerShell scripts, Visual and Android Studio
 - Developed .NET and Android web and mobile apps, and enhanced existing applications.
 - Maintained existing databases and developed stored procedures and alerts using T-SQL and PowerShell.
 - Provided customer service and support to company clients and internal users.
- May 2017 - April 2019
- RUTGERS UNIVERSITY | PISCATAWAY, NJ**
Software Developer, Unit Administrator (RUCDR Infinite Biologics - IT Department)
- Expert in C#, Visual Basic, .NET, JavaScript, Java, SQL server, MySQL and PostgreSQL, PHP, Laravel, HTML/CSS/JavaScript/jQuery, STARLIMS/SDMS, Crystal reports, Doctrine, Joomla
 - Conducted STARLIMS upgrade and participated in migrating its databases to the AWS platform.
 - Conducted upgrade of web apps to the latest versions of PHP and migrated their MySQL and Postgres DBs.

- Provided production support, enhancement, and maintenance for multiple databases, websites, and mobile apps; resolved and closed hundreds of reported bugs and tickets.

Jan 2016 -
April 2017

MEDLABS DIAGNOSTICS | CEDAR KNOLLS, NJ

.NET Developer (Department of IT)

- Specialist in C#, Visual Basic, .NET, SQL server 2012, SSIS/SSRS, HTML/CSS/JavaScript/jQuery
- Developed new and upgraded existing .NET web applications.
- Provided tech support and customer service on-site and remotely.
- Created software documentation for the newly created apps and updated existing documents.

Sept 2008 - May 2012

MOSCOW ACADEMY OF FINANCE AND LAW (MFUA)

Associate professor

- Taught undergraduate and graduate courses in Statistics, Financial Statistics, Budget system, Senior Project / Research (remotely since the summer 2010)
- Worked on several financial projects to stabilize the pension and social security systems of Russia.

Sept 2002 - May 2007

MIREA – RUSSIAN TECHNOLOGICAL UNIVERSITY (RTU)

Lecturer

- Taught undergraduate and graduate courses in Math Statistics / Statistics, Financial Mathematics, Insurance, Industrial Economics / Economics, and Budget system (developed)

EDUCATION

Jan 2029	PhD in Electrical and Computer Engineering – Machine Learning - RUTGERS UNIVERSITY NEW BRUNSWICK, NJ
May 2026	Master of Science in ECE – Machine Learning - RUTGERS UNIVERSITY NEW BRUNSWICK, NJ
Jan 2016	Master of Science in Computer Information Systems with Distinction - KEAN UNIVERSITY UNION, NJ
Jan 2015	Bachelor of Science in Computer Science, Summa Cum Laude – KEAN UNIVERSITY UNION, NJ
Jan 2015	Minor in Mathematics - KEAN UNIVERSITY UNION, NJ
Jan 2007	Doctor of Philosophy (Ph.D.) in Finance – INSTITUTE OF ECONOMICS, RUSSIAN ACADEMY OF SCIENCES (IERAS)
Jan 2002	Master of Science in Economics and Statistics - PLEKHANOV RUSSIAN UNIVERSITY OF ECONOMICS (MESI)
Jan 2001	Bachelor of Science in Economics and Statistics – PLEKHANOV RUSSIAN UNIVERSITY OF ECONOMICS (MESI)
Fellowship	NSF Computer and Information Science and Engineering Graduate Fellowship (CSGrad4US), Federal Award No. 2313998

MOST RECENT PUBLICATIONS in Computer Science and AI, * - in collaboration with Kean students

2026:

- [1] **Y. Kumar**, D. Kruger, J. J. Li, and S. Ghosh, “The LLM Story: A Data-Bound Visual Narrative for Transformer Education and Preliminary Diagnostics.” 2026 IEEE International Conference on Intelligent Systems and Interdisciplinary Applications (IEEE ISAIA 2026), Montclair, NJ, October 9–10, 2026 (accepted).
- [2] H. Rivera, **Y. Kumar**, D. Kruger, and J. J. Li, “Can LLMs Estimate Scale from 3D Meshes?” IEEE ISAIA 2026, Montclair, NJ, October 9–10, 2026 (accepted).
- [3] **Y. Kumar**, A. Luna, D. Kruger, J. J. Li, S. Sengupta, and H. Abdalla, “A Periodic Table of Data Poisoning.” IEEE ISAIA 2026, Montclair, NJ, October 9–10, 2026 (accepted).
- [4] D. Li, G. Yang, **Y. Kumar**, J. J. Li, and D. Kruger, “Auditing Occupational Score Stratification in Category-Visible LLM Resume Screening.” IEEE ISAIA 2026, Montclair, NJ, October 9–10, 2026 (accepted).
- [5] G. Yang, D. Li, **Y. Kumar**, D. Kruger, J. J. Li, H. Abdalla, and Y. S. Chung, “Optimization of Stability-Aware Self-Organizing Maps for Interdisciplinary Applications.” IEEE ISAIA 2026, Montclair, NJ, October 9–10, 2026 (accepted).
- [6] **Y. Kumar**, D. Kruger, J. J. Li, R. Thandra, S. Pandey, S. Guzman, J. Marchena, and S. Ghosh, “AutoNutriFact Agentic Lab: Scoring Evidential Truth in Diabetes Nutrition RAG Pipelines.” IEEE ISAIA 2026, Montclair, NJ, October 9–10, 2026 (accepted).
- [7] S. Ghosh, D. Kruger, J. J. Li, **Y. Kumar**, D. Sinnreich-Levi, K. August, and M. Striki, “Compiler-in-the-Loop Synthesis of GLSL Shaders and CMake Build Files with Local LLMs.” IEEE ISAIA 2026, Montclair, NJ, October 9–10, 2026 (accepted).

- [8] D. Kruger, S. Ghosh, **Y. Kumar**, D. Sinnreich-Levi, J. J. Li, K. August, M. Striki, and V. Lawrence, "AI Assisted Generation of Educational Software Components." IEEE ISAIA 2026, Montclair, NJ, October 9–10, 2026 (accepted).
- [9] D. Kruger, S. Ghosh, D. Sinnreich-Levi, M. Striki, J. J. Li, **Y. Kumar**, K. August, and V. Lawrence, "IDoc: Single-Source Authoring of Interactive, Accessible, Auto-Graded Documents." IEEE ISAIA 2026, Montclair, NJ, October 9–10, 2026 (accepted).
- [10] **Kumar, Y.**, Li, J.J., Kruger, D. (2026). A Systematic Analysis of AGI and ASI Existential Risk Scenarios. In: Arabnia, H.R., Deligiannidis, L., Amirian, S., Ghareh Mohammadi, F., Shenavarmasouleh, F. (eds) AI Revolution: Research, Ethics and Society. AIR-RES 2025. Communications in Computer and Information Science, vol 2721. Springer, Cham. https://doi.org/10.1007/978-3-032-12313-8_2.
- [11] D. Kruger, **Y. Kumar**, and J. J. Li, "Data Compression for AI Model Training," 2026 Data Compression Conference (DCC), p. 444, 2026, doi: 10.1109/DCC66757.2026.00071. <https://ieeexplore.ieee.org/document/11510384>
- [12] **Y. Kumar**, J. Mardi*, G. Yang, D. Kruger, and J. J. Li, "Preliminary Results of LLM Vulnerability Testing in Less Common Languages," in Applied Cognitive Computing and Artificial Intelligence, CSCE 2025, Communications in Computer and Information Science, vol. 2933, pp. 235–245, Springer, 2026. https://doi.org/10.1007/978-3-032-22205-3_17
- [13] Villalobos, W. *, **Y. Kumar**, D. Kruger, J. Jenny Li, Marchena, J. (2026). Collaborative Human–AI ageNt Swarm” for Conducting Scientific Research. 22nd International Conference on Intelligent Environments (*accepted*)
- [14] Loja J. *, **Kumar, Y.**, Li, J.J., Kruger, D. (2026). Human-Educator-In-The-Loop for Teaching with AutoML. 2026 IEEE Integrated STEM Education Conference (ISEC), Princeton, NJ, USA (*accepted*)
- [15] Hagg M. *, **Kumar, Y.**, Li, J.J., Kruger, D. (2026). Gamified Signal Processing Trainer with Interactive Visualizations and AI. 2026 IEEE Integrated STEM Education Conference (ISEC), Princeton, NJ, USA (*accepted*)
- [16] **Y. Kumar**, D. Lambropoulos, J. Hay, J. Cunha, K. Ponze, J. J. Li, and D. Kruger, "A Student-Friendly Quantum NLP Pipeline." 2026 IEEE Integrated STEM Education Conference (ISEC), Princeton, NJ, 2026 (*accepted*).
- [17] S. Ghosh, D. Kruger, M. Be, **Y. Kumar**, et al., "Kuiper TTS: Making Legacy Scientific Papers Accessible by Converting Equations into Spoken Language." 2026 IEEE Integrated STEM Education Conference (ISEC), Princeton, NJ, 2026 (*accepted*).
- [18] K. G. August, D. Kruger, D. Lambropoulos, T. M. Willis III, J. Komitas, J. J. Li, V. B. Lawrence, A. D. Dhillon, **Y. Kumar**, S. Ghosh, and M. Be, "Read the Math: Improving Mathematics Education Using Vision-Accessibility Techniques." 2026 IEEE Integrated STEM Education Conference (ISEC), Princeton, NJ, 2026 (*accepted*).
- [19] **Y. Kumar** and S. Sengupta, "Authenticating Matryoshka Nesting Dolls via an Auditable 2D–3D–Text Evidence Framework with BMA Compression and Zero-Shot 3D Completion," Electronics, vol. 15, no. 5, Art. no. 992, 2026. <https://doi.org/10.3390/electronics15050992>
- [20] **Y. Kumar** and S. Sengupta, "Authenticating Matryoshka Nesting Dolls via MML-LLM-Zero-shot 3D Reconstruction," Preprints.org, 2026 (preprint; superseded by the peer-reviewed Electronics article listed above). <https://www.preprints.org/manuscript/202602.0036>
- [21] H. B. Abdalla, **Y. Kumar**, J. J. Li, D. Kruger, and K. Ehimwenma, "Beyond Realism: A Utility-Fidelity-Privacy Framework for Trustworthy Synthetic Data," Journal of Data and Information Science, 2026. <https://doi.org/10.1515/jdis-2026-0038>
- [22] **Kumar, Y.**, G. Garcia-Vargas*, **J. J. Li**, D. Kruger, S. Aggarwal. (2026) Enhancing Airborne Surveillance with Live Feedback via Multimodal Skeleton-Based Pipeline. IEEE Access (*WIP*).
- [23] **Kumar, Y.**, Marchena, J. *, Kruger, D., & **Li, J. J.** (2026) A Multi-model Comparison of RAG Methods for Grant Requirement Extraction via SolicitationWizard. In *Intelligent Systems Conference*. Cham: Springer Nature Switzerland.
- [24] W. Villalobos, **Y. Kumar**, J. Marchena, J. J. Li, and D. Kruger, "Collaborative Human–AI ageNt Swarm for Conducting Scientific Research," in Proc. 20th International Conference on Digital Society (ICDS 2026), Venice, Italy, May 2026, pp. 8–15.
- [25] **Y. Kumar**, C. Paredes, M. Elhowary, D. Kruger, and J. J. Li, "Semi-Autonomous Harness DRAG for Repeatable Multilingual Robust Testing of LLMs," in Proc. 26th International Conference on Software Quality, Reliability and Security (QRS 2026), 2026, to appear.

2025:

- [24] **Kumar, Y.**, Thomas, S. *, Gayle, D. *, Li, J. J., & Kruger, D. (2025). AdversaGuard: A Distributed Data-Poisoning Benchmark for Parallel AI. https://sc25.supercomputing.org/proceedings/posters/poster_files/post110s2-file3.pdf
- [25] **Kumar, Y.**, Marchena, J. *, Guzman, S. *, Kruger, D., & Li, J. J. (2025, August). Comparative Analysis of RAG-Based Methods for Multilingual SwarmLexAI. In *Intelligent Systems Conference* (pp. 229-245). Cham: Springer Nature Switzerland. https://link.springer.com/chapter/10.1007/978-3-032-07109-5_16

- [26] Yang, G. *, Abdalla, H., **Kumar, Y.**, Kruger, D., & Li, J. J. (2025, August). Cross-Lingual and Multimodal Cyberbullying and Bias Detection and Content Generation via CyberGenDet. In Intelligent Systems Conference (pp. 210-224). Cham: Springer Nature Switzerland. https://link.springer.com/chapter/10.1007/978-3-031-99965-9_14
- [27] Kruger, D., **Kumar, Y.**, & Li, J. J. (2025, March). Parametric Matching for Improved Data Compression. In 2025 Data Compression Conference (DCC) (pp. 383-383). IEEE. <https://ieeexplore.ieee.org/abstract/document/10992580>.
- [28] Wong, G., **Kumar, Y.**, Li, J. J., & Kruger, D. (2025, April). Real-Time Object Detection and Skeletonization for Motion Prediction in Video Streaming. In Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 39, No. 28, pp. 29712-29714). <https://ojs.aaai.org/index.php/AAAI/article/view/35376>
- [29] **Y. Kumar**, B. Niu*, M. Lin* and N. Mudholker, "Text-to-Sign Language Video Generation Using GANs, BERT, and Sora," 2025 IEEE Integrated STEM Education Conference (ISEC), Princeton, NJ, USA, 2025, pp. 1-4, doi: 10.1109/ISEC64801.2025.11147381. <https://ieeexplore.ieee.org/document/11147381>
- [30] **Y. Kumar**, D. Kruger, J. Marchena* and J. J. Li, "Preliminary Results of Emotions Detection with AMIGO App," 2025 IEEE Integrated STEM Education Conference (ISEC), Princeton, NJ, USA, 2025, pp. 1-4, doi: 10.1109/ISEC64801.2025.11147309. <https://ieeexplore.ieee.org/document/11147309>
- [31] **Y. Kumar**, I. Akinwunmi* and D. Kruger, "Evaluating the Advantage of an AI-Native IDE Cursor on Programmer Performance," 2025 IEEE Integrated STEM Education Conference (ISEC), Princeton, NJ, USA, 2025, pp. 1-8, doi: 10.1109/ISEC64801.2025.11147402. <https://ieeexplore.ieee.org/document/11147402>
- [32] Jiang, R., **Kumar, Y.**, & Kruger, D. (2025). Enhanced Privacy-Preserving Architecture for Fundus Disease Diagnosis with Federated Learning. Applied Sciences, 15(6), 3004. <https://www.mdpi.com/2076-3417/15/6/3004>.
- [33] Abdalla, H. B., **Kumar, Y.**, Marchena, J. *, Guzman, S. *, Awlla, A., Gheisari, M., & Cheraghy, M. (2025). The Future of Artificial Intelligence in the Face of Data Scarcity. Computers, Materials & Continua, 84(1). <https://www.techscience.com/cmc/v84n1/61743>.
- [34] Abdalla, H. B., **Kumar, Y.**, Zhao, Y., & Tosi, D. (2025). A Comprehensive Survey of MapReduce Models for Processing Big Data. Big Data and Cognitive Computing, 9(4), 77. <https://www.mdpi.com/2504-2289/9/4/77>.

2024:

- [35] **Kumar, Y.**, Marchena, J. *, Awlla, A. H., **Li, J. J.**, & Abdalla, H. B. (2024). The AI-powered evolution of big data. Applied Sciences, 14(22), 10176. <https://www.mdpi.com/2076-3417/14/22/10176> (Currently with **115 citations.**)
- [36] **Kumar, Y.**, Lin, M. *, Paredes, C. *, Li, D. *, Yang, G. *, Kruger, D., Li, J. J., & Morreale, P. (2024). A Comprehensive Review of AI Advancement Using testFAILS and testFAILS-2 for the Pursuit of AGI. Electronics, 13(24), 4991. <https://doi.org/10.3390/electronics13244991>
- [37] **Kumar, Y.**, Paredes, C. *, Yang, G. *, Li, J. J., & Morreale, P. (2024, October). Adversarial Testing of LLMs Across Multiple Languages. In 2024 International Symposium on Networks, Computers and Communications (ISNCC) (pp. 1-6). IEEE. <https://ieeexplore.ieee.org/document/10758949>.
- [38] Ruiz, D. M. *, Watson, A. *, **Kumar, Y.**, Li, J. J., & Morreale, P. (2024, October). An AI-Powered Digital Foundation Recommender System. In 2024 International Symposium on Networks, Computers and Communications (ISNCC) (pp. 1-7). IEEE. <https://ieeexplore.ieee.org/document/10759049>.
- [39] **Kumar, Y.**, Huang, K., Perez, A. *, Yang, G. *, Li, J. J., Morreale, P., ... & Jiang, R. (2024). Bias and Cyberbullying Detection and Data Generation Using Transformer Artificial Intelligence Models and Top Large Language Models. Electronics, 13(17), 3431. <https://www.mdpi.com/2079-9292/13/17/3431>
- [40] Abdalla, H. B., Awlla, A. H., **Kumar, Y.**, & Cheraghy, M. (2024). Big Data: Past, Present, and Future Insights. In Proceedings of the 2024 Asia Pacific Conference on Computing Technologies, Communications and Networking (pp. 60-70). <https://dl.acm.org/doi/abs/10.1145/3685767.3685777>
- [41] Hannon, B. *, **Kumar, Y.**, Li, J. J., & Morreale, P. (2024). Chef Dalle: Transforming Cooking with Multi-Model Multimodal AI. Computers, 13(7), 156. <https://www.mdpi.com/2073-431X/13/7/156>
- [42] **Kumar, Y.**, Manikandan, A. *, Morreale, P., & Li, J. J. (2024, May). Growth Mindset Emojifier Multimodal App. In The International FLAIRS Conference Proceedings (Vol. 37). <https://journals.flvc.org/FLAIRS/article/view/135572>
- [43] **Kumar, Y.**, Huang, K., Lin, C. C. *, Watson, A. *, Li, J. J., Morreale, P., & Delgado, J. * (2024). Applying Swin Architecture to diverse Sign Language Datasets. Electronics, 13(8), 1509. <https://www.mdpi.com/2079-9292/13/8/1509>
- [44] **Kumar, Y.**, Manikandan, A. *, Kupershtein, E. *, Li, J. J., & Morreale, P. (2024, March). Assessing the Impact of Professional Development on K-12 CS Education: A One-Year Follow-Up Survey Analysis. In 2024 IEEE Integrated STEM Education Conference (ISEC) (pp. 1-8). IEEE. <https://ieeexplore.ieee.org/abstract/document/10665196>
- [45] **Kumar, Y.**, Manikandan, A. *, Li, J. J., & Morreale, P. (2024, March). Optimizing Large Language Models for Auto-Generation of Programming Quizzes. In 2024 IEEE Integrated STEM Education Conference (ISEC) (pp. 1-5). IEEE. <https://ieeexplore.ieee.org/abstract/document/10665141>

- [46] **Kumar, Y.**, Manikandan, A. *, Li, J. J., & Morreale, P. (2024). Preliminary Results from Integrating Chatbots and Low-Code AI in CS Coursework. In 2024 IEEE Integrated STEM Education Conference (ISEC) (pp. 1-4). IEEE. <https://ieeexplore.ieee.org/abstract/document/10665039>
- [47] Hannon, B. *, **Kumar, Y.**, Gayle, D. *, Li, J. J., & Morreale, P. (2024). Robust Testing of AI Language Model Resiliency with Novel Adversarial Prompts. *Electronics*, 13(5), 842. <https://www.mdpi.com/2079-9292/13/5/842>
- [48] Villalobos, W. *, **Kumar, Y.**, & Li, J. J. (2024). The Multilingual Eyes Multimodal Traveler's App. In International Congress on Information and Communication Technology (pp. 565-575). Singapore: Springer Nature Singapore. https://link.springer.com/chapter/10.1007/978-981-97-3305-7_45
- [49] **Kumar, Y.**, Gordon, Z. *, Alabi, O. *, Li, J., Leonard, K., Ness, L., & Morreale, P. (2024). ChatGPT Translation of Program Code for Image Sketch Abstraction. *Applied Sciences*, 14(3), 992. <https://www.mdpi.com/2076-3417/14/3/992>
- [50] **Kumar, Y.**, Huang, K., Gordon, Z. *, Castro, L. *, Okumu, E. *, Morreale, P., & Li, J. J. (2024). Transformers and LLMs as the New Benchmark in Early Cancer Detection. In ITM Web of Conferences (Vol. 60, p. 00004). EDP Sciences. https://www.itm-conferences.org/articles/itmconf/abs/2024/03/itmconf_aiss2024_00004/itmconf_aiss2024_00004.html

2023:

- [51] **Kumar, Y.**, Delgado, J. *, Kupershtein, E. *, Hannon, B. *, Gordon, Z. *, Li, J. J., & Morreale, P. (2023, October). AssureAIDoctor-A Bias-Free AI Bot. In 2023 International Symposium on Networks, Computers and Communications (ISNCC) (pp. 1-6). IEEE. <https://ieeexplore.ieee.org/abstract/document/10323978>
- [52] **Kumar, Y.**, Morreale P, Sorial P *, Delgado J *, Li JJ, Martins P. *. A Testing Framework for AI Linguistic Systems (testFAILS). *Electronics*. 2023; 12(14):3095. <https://www.mdpi.com/2079-9292/12/14/3095>
- [53] A. Balcacer *, B. Hannon *, **Y. Kumar**, K. Huang, J. Sarnoski, S. Liu, J. J. Li, P. Morreale (2023) Mechanics of a Drone-Based System for Algal Bloom Detection Utilizing Deep Learning and LLMs (MIT URTC 2023) <https://ieeexplore.ieee.org/abstract/document/10534955>
- [54] **Y. Kumar**, P. Morreale, P. Sorial *, J. Delgado *, J. J. Li and P. Martins, "A Testing Framework for AI Linguistic Systems (testFAILS)," 2023 IEEE International Conference On Artificial Intelligence Testing (AITest), Athens, Greece, 2023, pp. 51-54, doi: 10.1109/AITest58265.2023.00017, <https://ieeexplore.ieee.org/document/10229447>
- [55] Hannon, B. *, **Kumar, Y.**, Sorial, P. *, Li, J. J., & Morreale, P. (2023, May). From Vulnerabilities to Improvements: A Deep Dive into Adversarial Testing of AI Models. In Proceedings of the 21st International Conference on Software Engineering Research & Practice (SERP) (pp. 24-27). <https://ieeexplore.ieee.org/abstract/document/10487199>
- [56] Kupershtein, E. *, **Kumar, Y.**, Manikandan, A. *, Morreale, P., & Li, J. J. (2023, July). ChatGPT: A Game-Changer for Embedding Emojis in Faculty Feedback. In Proceedings of the 19th International Conference on Frontiers in Education: Computer Science & Computer Engineering (FECS). <https://ieeexplore.ieee.org/abstract/document/10487568>
- [57] **Y. Kumar**, W. Li *, K. Huang, M. Thompson *, B. Hannon * (2023) Natural Language Coding (NLC) for Autonomous Stock Trading: A New Dimension in No-Code/Low-Code (NCLC) AI. (QRS 2023) <https://ieeexplore.ieee.org/abstract/document/10430057>
- [58] **Y. Kumar**, Z. Gordon *, P. Morreale, J. Jenny Li, B. Hannon *. (2023) Love the Way You Lie: Unmasking the Deceptions of LLMs. (QRS 2023) <https://ieeexplore.ieee.org/abstract/document/10429948>
- [59] Chu, J., **Kumar, Y.**, Kwak, D., Novotny, J., Patel, P., & Morreale, P. (2023, March). Evaluating the Effectiveness of Equitable K-12 Professional Learning Access in Computer Science. In 2023 IEEE Integrated STEM Education Conference (ISEC) (pp. 222-227). IEEE.

2022:

- [60] J. Delgado *, U. Ebrese *, **Y. Kumar**, J. J. Li and P. Morreale, "Preliminary Results of Applying Transformers to Geoscience and Earth Science Data," 2022 International Conference on Computational Science and Computational Intelligence (CSCI), Las Vegas, NV, USA, 2022, pp. 284-288, doi: 10.1109/CSCI58124.2022.00054.
- [61] Tellez, N. *, Serra, J. *, **Kumar, Y.**, Li, J.J., Morreale, P. (2023). Gauging Biases in Various Deep Learning AI Models. In: Arai, K. (eds) Intelligent Systems and Applications. IntelliSys 2022. Lecture Notes in Networks and Systems, vol 544. Springer, Cham.
- [62] N. Tellez *, J. Serra *, **Y. Kumar**, J. J. Li, P. Morreale (2022). "An Assure AI Bot (AAAI bot)," 2022 International Symposium on Networks, Computers and Communications (ISNCC), 2022, pp. 1-5, doi: 10.1109/ISNCC55209.2022.9851759.
- [63] J. Serra *, S. Fortes *, A. Allaico *, E. Landaverde *, R. Quezada *, **Y. Kumar**, J. J. Li, P. Morreale, 2022. Validation of AI models for ITCZ Detection from Climate Data. In Proceedings of International Conference on Data Science and Information Technology 5th International Conference on Data Science and Information Technology (DSIT), 2022, pp. 1-7, doi: 10.1109/DSIT55514.2022.9943879

- [64] Patel, P. *, Morreale, P., **Kumar, Y.**, Kwak, D., Chu, J., Garcia, R., & Burnett, M. (2022, March). Embedding Equitable Design in the CS Computing Curricula. In Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 2 (pp. 1332-1332).
- [65] Patel, P. *, Chu, J., **Kumar, Y.**, Kwak, D., Morreale, P., Garcia, R., & Burnett, M. (2022, March). Implementing Inclusive Software Design in the CS Curriculum. In Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 2 (pp. 1272-1272).
- [66] Daehan Kwak, Patricia Morreale, Sarah T. Hug, **Kumar, Y.**, Jean Chu, Ching-Yu Huang, J. Jenny Li, Paoline Wang. Evaluation of the Use of Growth Mindset in the CS Classroom: Proceedings of the 53rd ACM Technical Symposium on Computer Science Education (SIGCSE'22), V. 1, February 2022, Pages 878–884.
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Earlier Works: †-former name

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